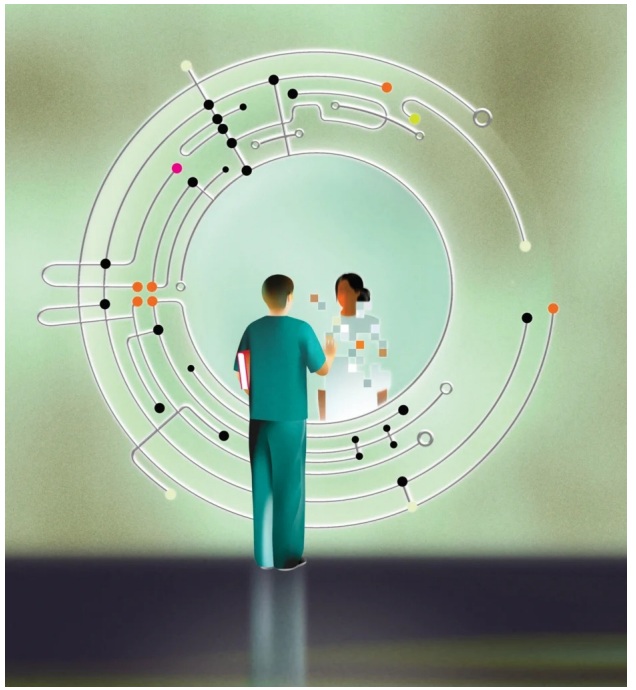
The University of Texas at Austin
Dell Medical School

JOINT TEXAS PEM FELLOWSHIP CONFERENCE · NOVEMBER 11, 2026

LEARNING WITH AI: FROM CLASSROOM TO BEDSIDE



AI in Medical Education

NIHARIKA GOPARAJU, MD, MPH
Assistant Professor - Pediatric Emergency Medicine Faculty - UT Austin Dell Medical School – Dell Children's Medical Center

DISCLOSURES

I have no relevant financial relationships to disclose

Financial relationships	None
Products named	Tools are named as examples only, not endorsements.
AI use in this talk	AI assisted with literature search, creating infographics, drafting the slide design. I reviewed and verified all content and references.

I KNOW. ANOTHER AI TALK.

So let's start with the real debate: AI in education

SCHOOL OF THOUGHT 1

Restrict to protect

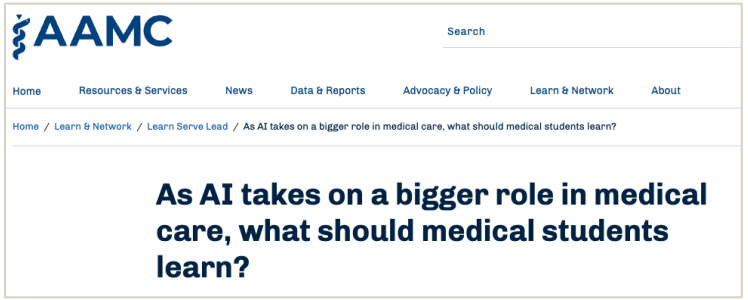
SCHOOL OF THOUGHT 2

Teach to upskill

HEADLINES







TWO DISTINCT APPROACHES

Teaching medicine in an AI era vs. teaching clinicians about using AI.

Dimension	Teaching medicine in the AI era	Teaching learners about AI
Core question	*"What should we teach when AI can diagnose and write prescriptions?"*	*"How do we safely choose, operate, audit, and govern AI tools?"*
Primary goal	Rebalance medical training from recall to human-centric bedside skills and reasoning.	Build AI literacy, error detection, and calibrated skepticism.
Educational focus	AI-free zones, formative clinical struggle, empathy, communication, and executive oversight.	Choosing tools, prompting, hallucination and bias recognition, and data ethics.



By the way, ChatGPT misspelled your name.

LEARNING OBJECTIVES

By the end of this session, you will be able to:

1 **Describe how medical students and schools use AI in education today**
Section 1 · Current AI use in education

2 **Identify the risks and opportunities AI creates for teaching medicine**
Section 2 · Teaching medicine in the AI era

3 **Outline the AI literacy skills our learners need**
Section 3 · Teaching learners about AI

4 **Apply AI tools to everyday PEM teaching and learning**
Section 4 · Practical applications and demonstrations

Niharika's website and AI literacy resources

niharikagoparaju.github.io/ai-literacy

SECTION 1

Current AI Use in Education

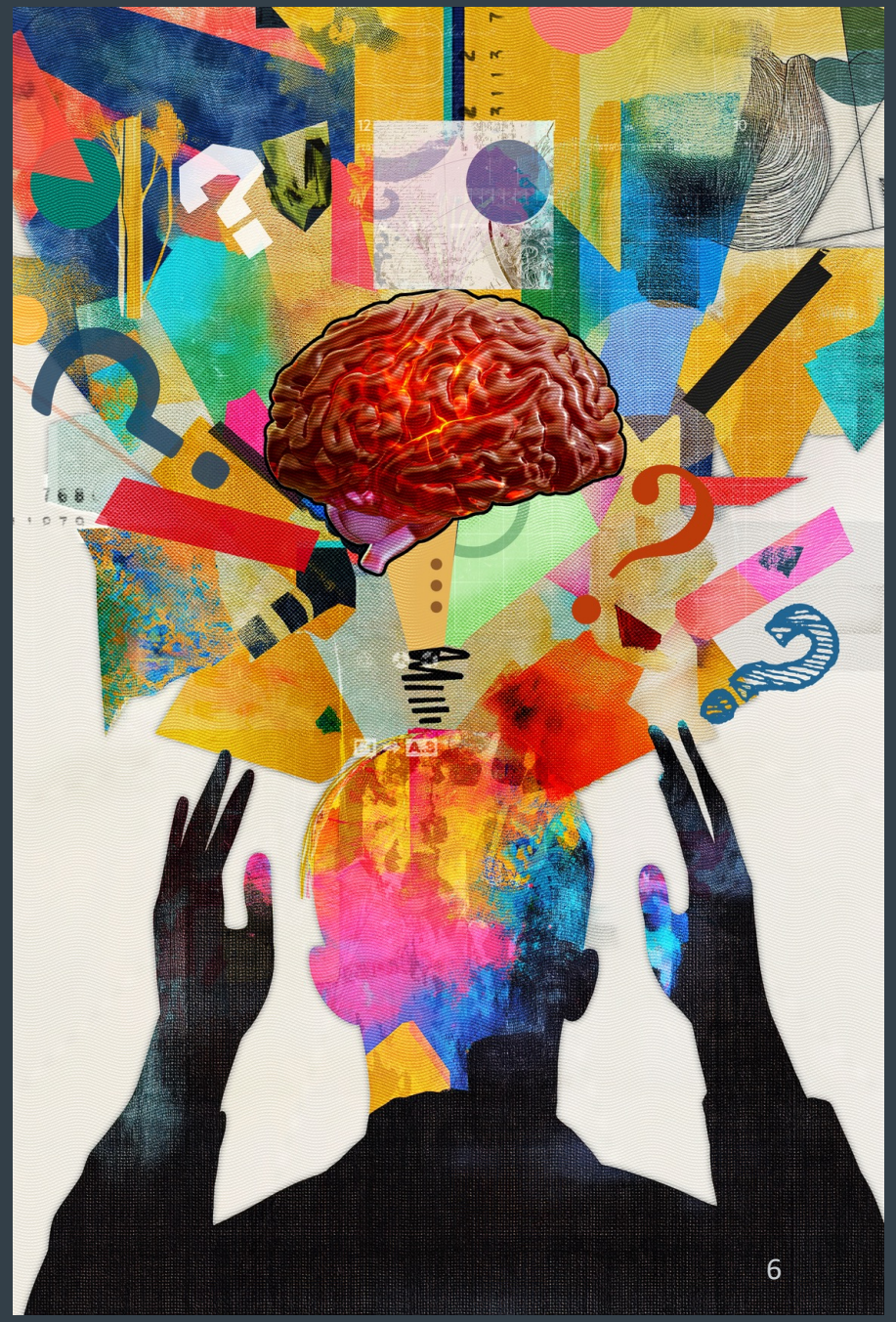


Image: Getty Images

SECTION 1 · CURRENT AI USE IN EDUCATION

Most K-12 students now say their school has AI rules and allows AI for schoolwork

FIGURE 14

Gen Z K-12 Students' Reports of Artificial Intelligence Policy and Access

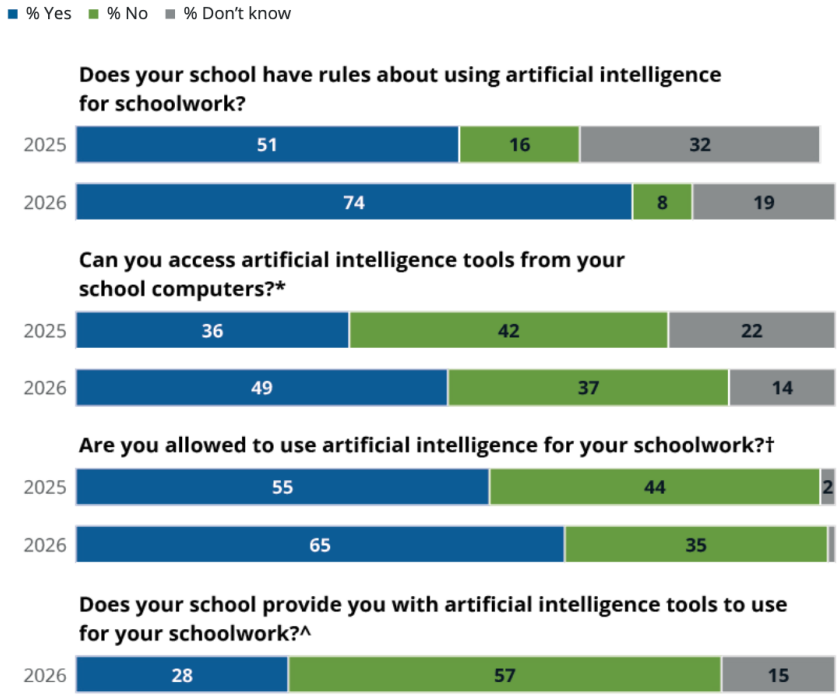
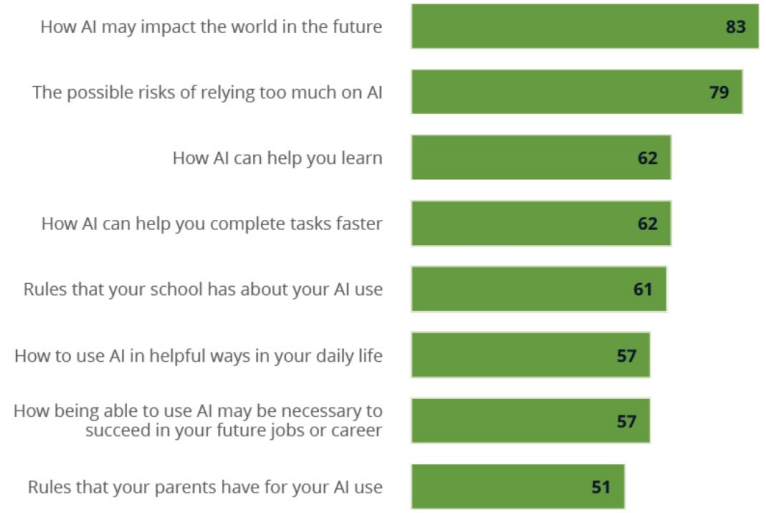


FIGURE 16

Gen Z K-12 Students' Conversations With Parents About Artificial Intelligence

When you talked to your parents about AI, did you talk to them about the following?

% Yes



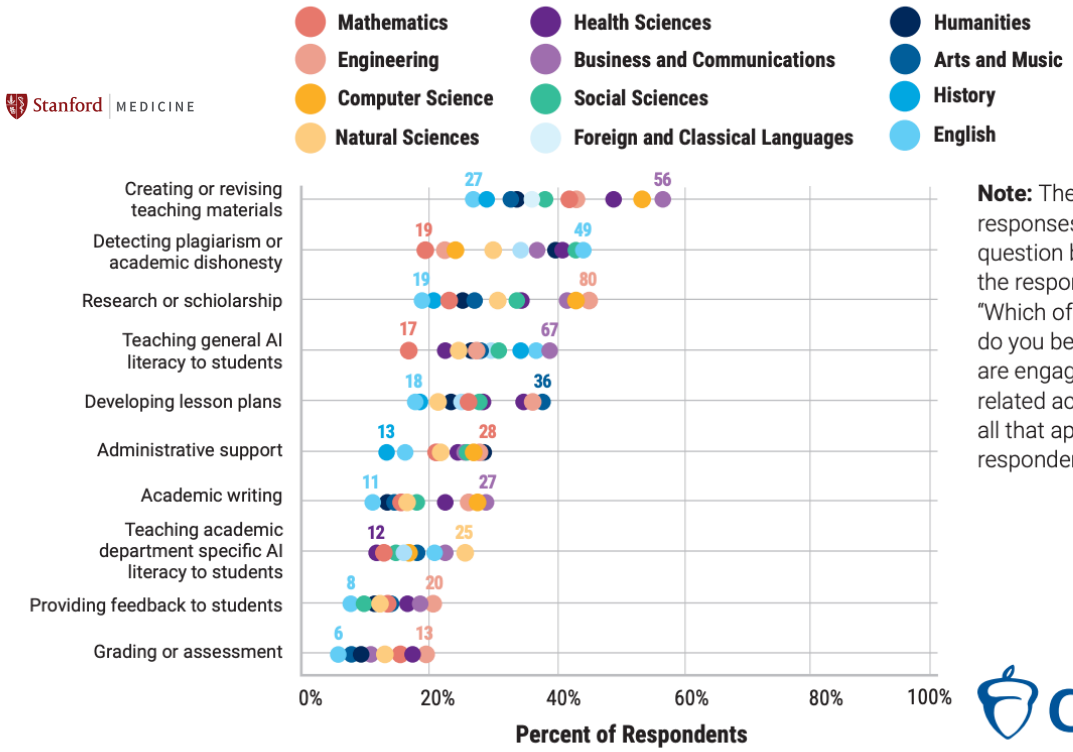
Note: Among the 59% of Gen Z K-12 students who have ever talked to their parents about AI

Source: Walton Family Foundation, GSV Ventures, Gallup. Voices of Gen Z: The AI Paradox, More Exposure, Less Confidence Among Gen Z. 2026. Figures 14 and 16.

SECTION 1 · CURRENT AI USE IN EDUCATION

Faculty use of AI varies widely by discipline

Figure 7: Faculty Uses of AI, by Faculty Academic Discipline



Note: The figure reports responses to the following question by the discipline of the respondent's department: "Which of the following AI uses do you believe your students are engaging in for classroom-related activities? Please select all that apply." Total number of respondents = 2,988.

CollegeBoard

Source: College Board faculty survey, Figure 7. Faculty reports of AI use for work activities; n=2,988.

MEDICAL SCHOOL

How medical schools are approaching incorporating AI

GUARDED — EMBEDDED

School	Approach	What they are doing
Stanford	Embedded	AI literacy modules and workshops, approved study tools and patient sims for rotations.
NYU Grossman	Embedded	Precision education; AI feedback on notes and communication
Harvard	Targeted	AI course for incoming HST students; AI in Medicine PhD track
Mount Sinai · Hopkins · UCSF · Cincinnati	Guided tools	Study tools, secure ChatGPT Edu, note feedback, and a Socratic coach.
Dell Med	Developing	Providing tools, AI in specific lectures, a healthcare AI seminar series, recent UT Real Health AI symposium

Stanford Medicine Magazine 2025 · Mount Sinai 2025 · AAMCNews 2025–26 · HMS 2024 · Jamieson AR, et al. NEJM AI 2024 · Kulkarni-Date M, Bixby J 2026

MEDICAL SCHOOL

Lots of AI medical education research

Macy Foundation Innovation Report Part I: Current Landscape of Artificial Intelligence in Medical Education

Christy K. Boscardin, PhD, Raja-Elie E. Abdulnour, MD, and Brian C. Gin, MD, PhD

Table 1
Education-Related Activities and the AI Tasks That Support Them[a]

Domain	AI tasks: Knowledge retrieval and interaction (LLMs ± RAG)	Classifiers and predictors (LLMs, NNs, Trad ML)	Summarizer (LLMs, speech recognition)	Virtual agent (LLMs)	Machine vision and spatial reasoning (NNs, LLMs, diffusion)
Admissions	N = 0	N = 16 • Identifying themes and biases in LORs and personal statements • Applications screening/ranking	N = 3 • Generating LOR and personal statements	N = 2 • Informational chatbot	N = 0
Classroom-based learning and teaching	N = 82 • Knowledge reference • Generating learning cases • Clinical reasoning support • Personalized curricula	N = 73 • Learning needs identification • Further reading suggestion	N = 2 • Generate study aids, flashcards	N = 10 • Communications simulation, virtual patient • Virtual tutor	N = 18 • Procedure simulation • Generating images of physical findings
Workplace-based learning and practice	N = 41 • Knowledge reference • CDS, DDS • Feedback on documentation	N = 84 • Documentation feedback • Patient care analytics	N = 5 • AI scribe (dictation, documentation) • Case consolidation and logging, tracking clinical experiences	N = 12 • Virtual coach for clinical decision making	N = 41 • Radiology, pathology, cardiology, dermatology interpretation • Procedure guidance (including robotic surgery)
Assessment, feedback, and certification	N = 133 • Generating assessment content and answer keys: MCQs, OEQs, cases • Generating personalized exam feedback	N = 165 • Automated scoring • Performance prediction • Competency categorization • Narrative feedback themes	N = 9 • Summarize narrative assessments and feedback • Performance dashboards • Evaluating student self-assessments	N = 16 • Virtual standardized patients for OSCEs and other assessments	N = 33 • Procedural skills assessment • Eye tracking analysis
Program evaluation and research	N = 18 • Evaluating/auditing curricula, courses, and clinical rotations • Literature exploration	N = 21 • Coding and thematic analysis • Identification of bias • Predicting risk for burnout • Predicting career trends	N = 4 • Evaluating and mapping curricular content	N = 0	N = 1 • Assessing patient diversity in teaching materials

Abbreviations: AI, artificial intelligence; CDS, clinical decision support; DDS, diagnostic support; LLM, large language model; LOR, letter of recommendation; MCQs, multiple choice questions; NN, neural network; OEQs, open-ended questions; OSCEs, objective structured clinical exams; RAG, retrieval-augmented generation; Trad ML, traditional machine learning.

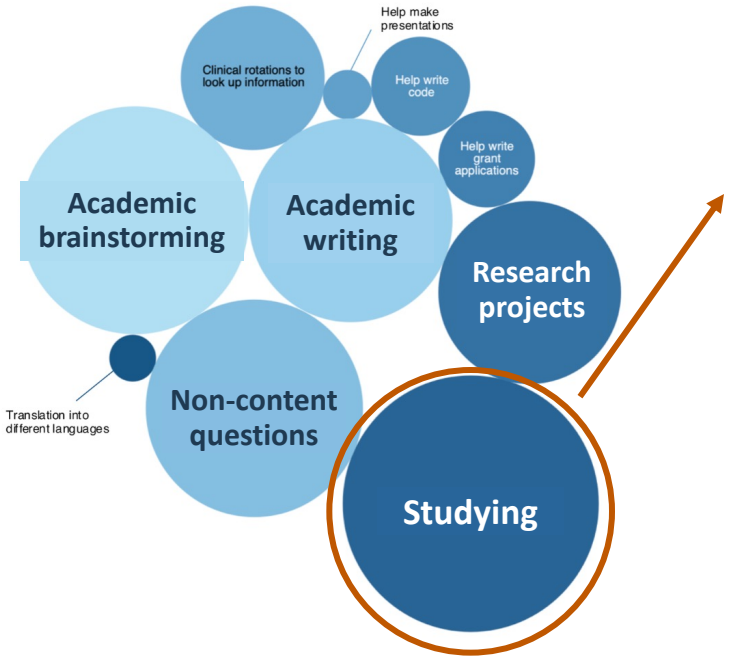
[a]N indicates the number of reviewed articles that relate to the bulleted activities.

Domains: Boscardin CK, et al. Acad Med 2025 (Macy Foundation report).

MEDICAL SCHOOL

Half of U.S. medical students already use ChatGPT for coursework, most often to study.

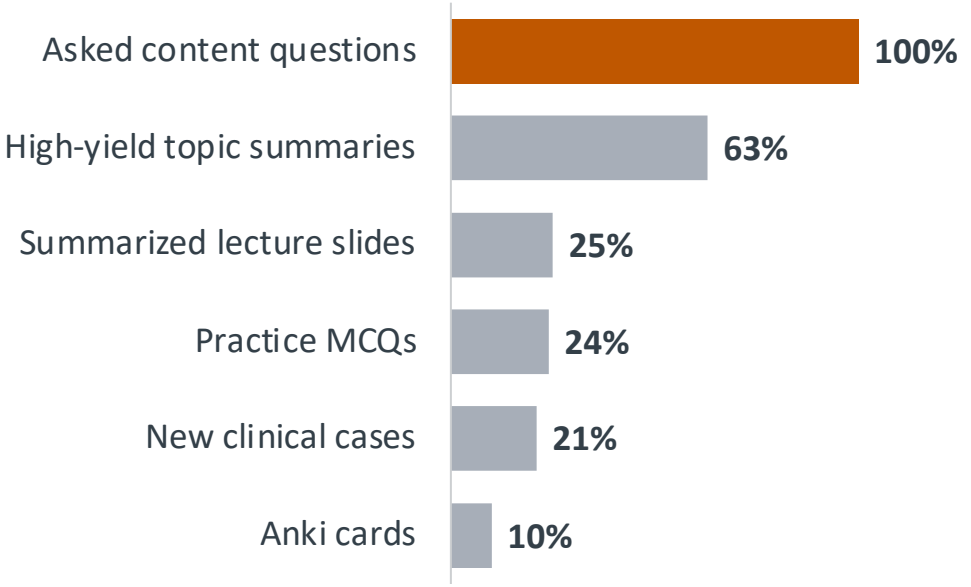
What students used ChatGPT for



38%

of students used ChatGPT to study

How they studied with it, % of that 38%



Source: Ganjavi C, et al. PLOS Digital Health, 2024; survey of 415 U.S. medical students at 28+ schools, May 2023.

SECTION 1 · CURRENT AI USE IN EDUCATION

There is no shared AI curriculum; each school is building its own

WHAT SCHOOLS ARE DOING

- **Required modules + workshops**
- **Clinical Tools**
- **Study tools + secure ChatGPT**
- **Lectures, seminars, degrees (optional)**

WHAT NATIONAL BODIES OFFER

AAMC Principles (2025)
Responsible-use guidance

AAMC AI Competencies (2026)
26 competencies; a target, not a course

ACGME (residency)
No AI requirements yet

AAP · ACEP · SAEM (2025–26)
Call for AI literacy; no curriculum

Still debated: What belongs in it? How often should it be updated?

Sources: AAMC Principles (2025); AAMC AI Competencies (2026); AAP policy statement, Pediatrics (Nov 2026); ACEP AI Task Force consensus (2025); SAEM AI Consensus Conference (2026); school websites.

SECTION 2

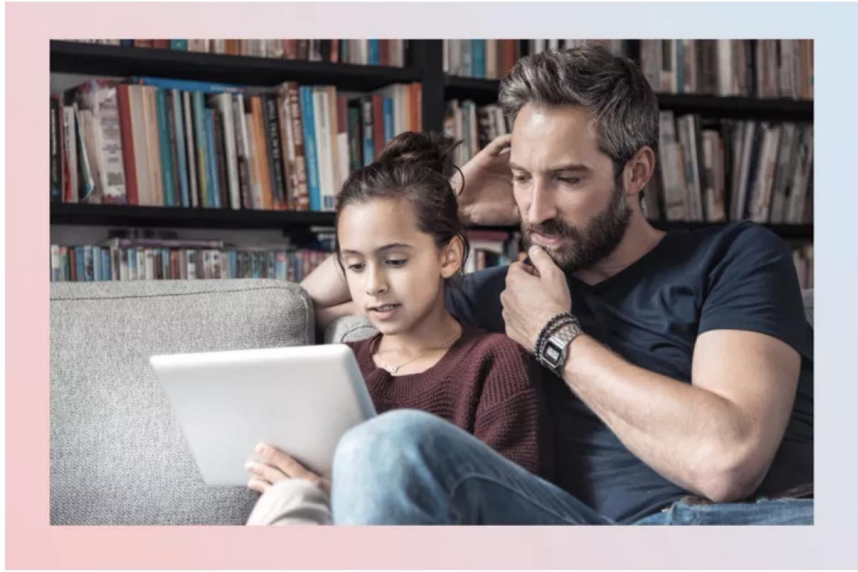
Teaching Medicine in the AI Era

NEWS

New Study Says Parents Trust ChatGPT for Health Advice Over Doctors

Experts say this is convenient but risky.

By **Sherri Gordon, CLC** | Published on October 18, 2024

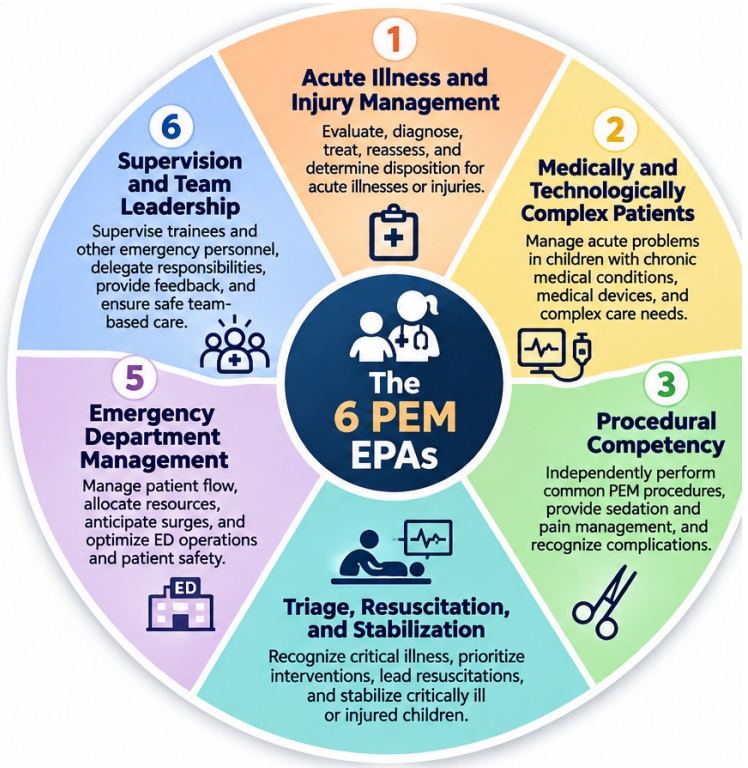


GETTY IMAGES/GOLERO

Research from the University of Kansas Life Span Institute found that parents seeking health care information online for their children trust artificial intelligence (AI) like ChatGPT more than health care

ChatGPT was rated as more trustworthy, accurate, and reliable.

The six PEM EPAs as our framework



EPA & Topic	Essential Soft Skills
EPA 1: Acute Illness and Injury	Family-centered communication; Explaining diagnoses; Shared decision-making; Obtaining informed consent
EPA 2: Medically Complex Patients	Empathy; Communicating with caregivers; Managing uncertainty; Interdisciplinary collaboration
EPA 3: Procedural Competency	Explaining risks and benefits; Obtaining consent; Reassuring patients and families
EPA 4: Resuscitation and Stabilization	Team leadership; Closed-loop communication; Coordination under extreme pressure
EPA 5: Emergency Department Management	Teamwork; Prioritization; Adaptability; Effective communication during high-stress situations
EPA 6: Supervision and Leadership	Teaching; Delegation; Conflict resolution; Feedback; Managing complaints, service recovery

Gemini Notebook

Underlying these EPAs are skills we need to **emphasize even more in the AI era**

American Board of Pediatrics, PEM EPAs · Hsu DC, et al. Pediatr Emerg Care 2023;39(8):574-579 · Wheel: speaker's infographic

SECTION 2 · TEACHING MEDICINE IN THE AI ERA

Patients and parents value these as much as competence

WHAT PATIENTS AND PARENTS VALUE

Empathy

Communication skills

Partnership

Thoroughness

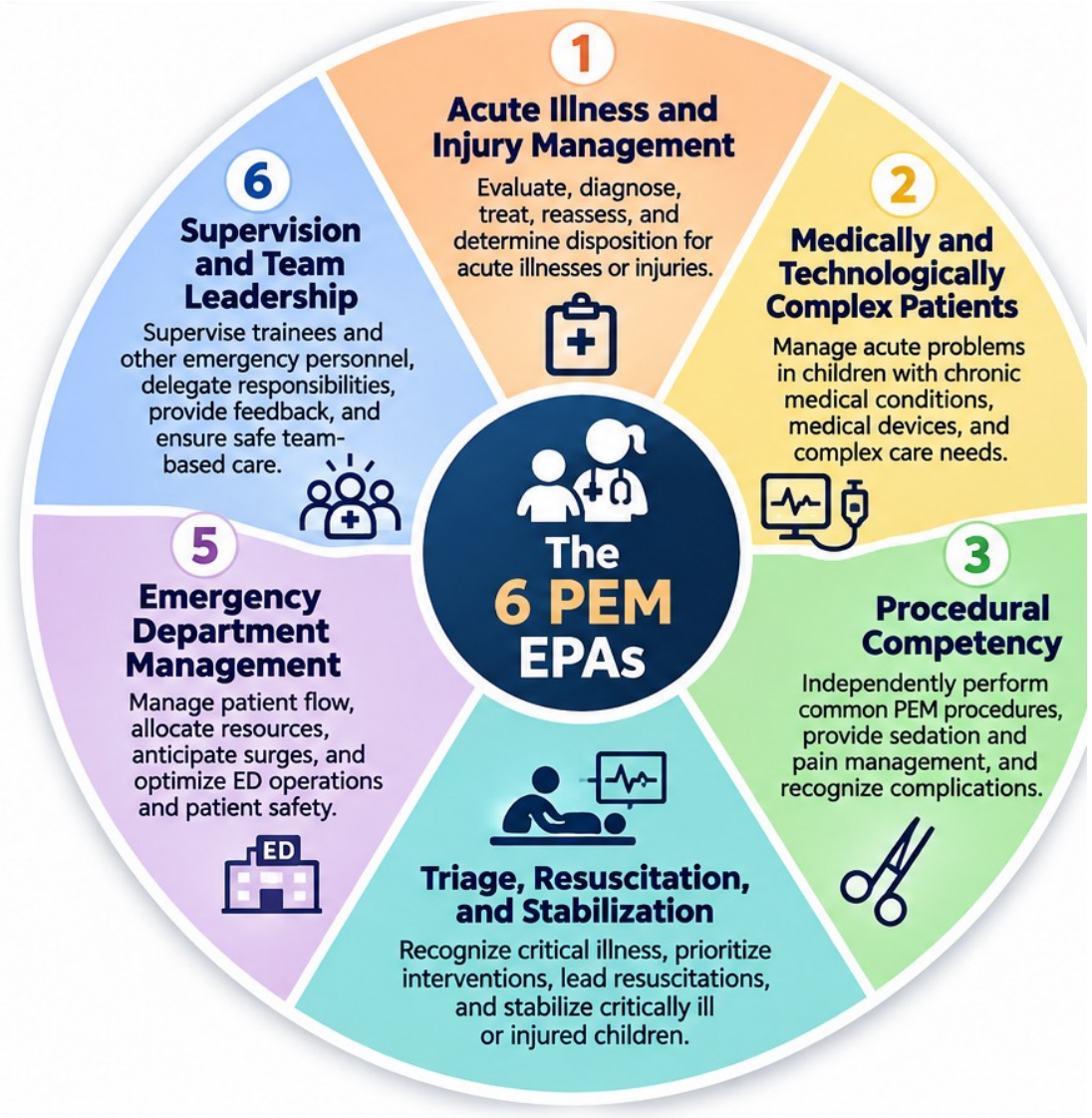


Luke Fildes, *The Doctor*, 1891, oil on canvas

1 “People choose to receive human empathy despite rating AI empathy higher” (PMC12872445). 2 Pew Research Center, n=3,488 U.S. adults, June 2026. 3 JAMA Netw Open, Oct 2026; n=1,185 U.S. adults. 4 BMC Health Serv Res 2025 (CCFNI-ED survey of PED parents).

SECTION 2 · TEACHING MEDICINE IN THE AI ERA

The six PEM EPAs as our framework



CENTRAL QUESTION

How can AI help us teach or learn the activities PEM fellows must eventually perform *independently*?

American Board of Pediatrics, PEM EPAs · Hsu DC, et al. Pediatr Emerg Care 2023;39(8):574-579 · Wheel: speaker's infographic

SECTION 2 · TEACHING MEDICINE IN THE AI ERA

AI may help train certain clinical tasks better than others

PEM EPA	AI EDUCATIONAL FIT	EXAMPLE USE IN TRAINING
1 **Acute Illness and Injury Management**	Moderate	Fellowship: didactics - Case practice: differentials, plans, feedback on reasoning.
2 **Medically and Technologically Complex Patients**	Moderate	Fellowship didactics: review of rare conditions, devices, and care plans
3 **Procedural Competency**	Limited	Pre-procedure review only; the skill is built by doing
4 **Triage, Resuscitation, and Stabilization**	Limited	Sim scenario writing; resuscitation is learned hands-on
5 **Emergency Department Management**	Limited	Tabletop surge and patient-flow scenarios
6 **Supervision and Team Leadership**	Limited	Rehearse feedback conversations and teaching scripts

EPAs: American Board of Pediatrics; Hsu DC, et al. Pediatr Emerg Care 2023 · Fit ratings: speaker's assessment

SECTION 2 · TEACHING MEDICINE IN THE AI ERA

What techniques are proven to help with learning?

THREE METHODS WITH STRONG EVIDENCE

Retrieval practice

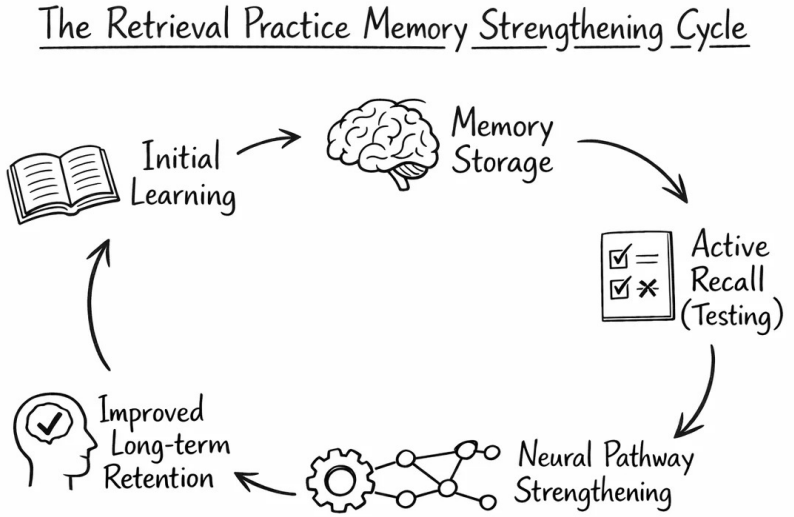
Pull it from memory instead of rereading it

Spacing

Revisit a topic over days and weeks

Deliberate practice

Targeted reps on weak areas, with feedback



Content outline: American Board of Pediatrics, Pediatric Emergency Medicine content outline. Diagram: Structural Learning.

SECTION 2 · TEACHING MEDICINE IN THE AI ERA

How Can AI Help?

HOW AI SUPPORTS THE EDUCATOR · RECLAIMING FACULTY TIME

AI Can Draft Content and Help With Planning

Board-style question or games
Item drafts mapped to the PEM content outline

Curriculum mapping
Sessions and cases checked against the six PEM EPAs

Simulation case drafts
Scenario, vitals and critical actions for faculty to revise

OUR PROGRAM, FOR EXAMPLE

Curriculum workshop: revamped our fellowship curriculum

1. Needs assessment survey of fellows
2. Combined national requirements, our goals, and recent ITE results
3. Used Gemini to map gaps and draft the new curriculum while having a 2-day in person workshop where faculty verify/add.
4. Cut low-yield content; added simulation

[1]Anthropic Education Report: How educators use Claude, 2025 (74,000 conversations). [2]Frontiers in Medicine. 2026;13:1978708 (GPT-5.6, Claude-5, Gemini-3.5 and others). [3]medRxiv preprint, 2026 (GPT-5.1/5.2, Gemini 3.0 Pro, Claude Opus 4.5).

SECTION 2 · TEACHING MEDICINE IN THE AI ERA

Two 2026 studies tested current models on teaching content

CURRENT-MODEL EVIDENCE · RESEARCH QUESTIONS AND SETTING

Study	Board-style questions	Pediatric OSCE stations
Research question	Can current models write epilepsy multiple-choice questions ready for teaching?	How much do current models differ when writing the same pediatric OSCE station?
Models	GPT-5.6, Claude-5, Gemini-3.5 and four others	GPT-5.1, GPT-5.2, Gemini 3.0 Pro, Claude Opus 4.5, Grok 4.1
What they wrote	60 questions per model, 420 in total	6 neonatal jaundice stations per model, 30 in total
Reviewed by	Two blinded clinicians, six areas scored 1–5	Four blinded experts, including two pediatricians
Published	Frontiers in Medicine, Oct 2026	medRxiv preprint (not peer reviewed), Aug 2026

Sources: Leveraging LLMs to generate MCQs for epilepsy education. Front Med. 2026;13:1978708. Characterizing LLM variability in OSCE stations. medRxiv. 2026. doi:10.64898/2026.08.04.26359691

SECTION 2 · TEACHING MEDICINE IN THE AI ERA

Current models still make errors and need expert review

CURRENT-MODEL EVIDENCE · KEY FINDINGS

Finding	Board-style questions	Pediatric OSCE stations
Looked right	All 420 met the required format after one correction	Followed the prompt; few factual errors
Errors	Wrong answer key in 2–38% of questions, by model	Only 1 of 5 models covered all three types of jaundice
Weakest area	Wrong options too easy to rule out (2–3 of 5)	No model covered every communication skill
Ready to use	0.5% usable as written; 71% needed major revision	No model ready for unsupervised use in an exam bank

> **Both author teams:** use these models to draft, with clinical review before anything reaches learners.

Sources: Front Med. 2026;13:1978708 (two blinded reviewers). medRxiv preprint, 2026, doi:10.64898/2026.08.04.26359691 (four blinded raters).

SECTION 3

Teaching Learners About AI

SECTION 3 · TEACHING LEARNERS ABOUT AI

What should we teach?

TRACK 1 · EVERYDAY USE

Use AI without losing clinical skills

Who: Every clinician and trainee

- Know what the tool does well and poorly
- Use AI to review and support your management vs. replace it

TRACK 2 · EVALUATION AND ADOPTION

Evaluating AI Tools

Who: Clinical leaders, Informatics, AI researchers

- Appraise AI research and vendor claims
- Check validation, bias and fit for children
- Weigh automation risks in the workflow
- Know where to keep learning as tools change

SECTION 3 · TEACHING LEARNERS ABOUT AI

Three ways reliance on AI can shape clinical skill

DEFINITIONS · ABDULNOUR ET AL., NEJM 2025

Deskilling

A skill the clinician already had erodes from disuse.

Never-skilling

A foundational skill never develops because AI did the task first.

Mis-skilling

The learner adopts the AI's errors or biases as their own reasoning.

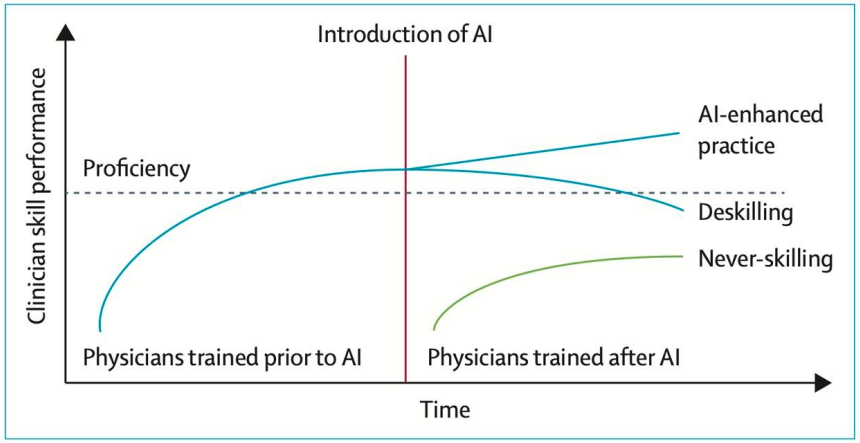
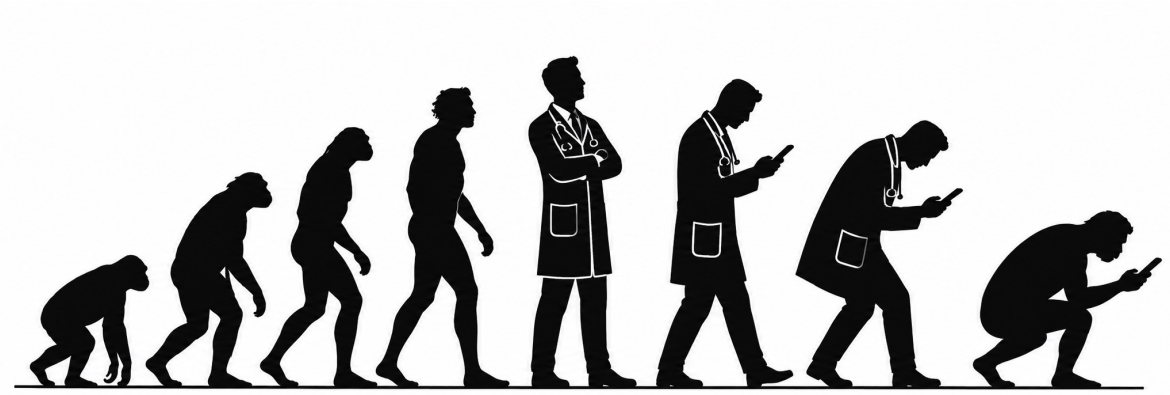


Figure: **Potential effects of AI on clinical skills over time**



Source: Abdulnour REE, Gin B, Boscardin CK. N Engl J Med. 2025;393:786-797. Examples are illustrative.

SECTION 3 · TEACHING LEARNERS ABOUT AI

Redesigning clinical reasoning education

SCHAYE ET AL. · ABOUT THE PAPER

The paper

Theory-informed redesign of clinical reasoning education in the era of generative artificial intelligence

Schaye et al. · Academic Medicine, 2026 · Scholarly Perspective

Proposes five cross-cutting strategies

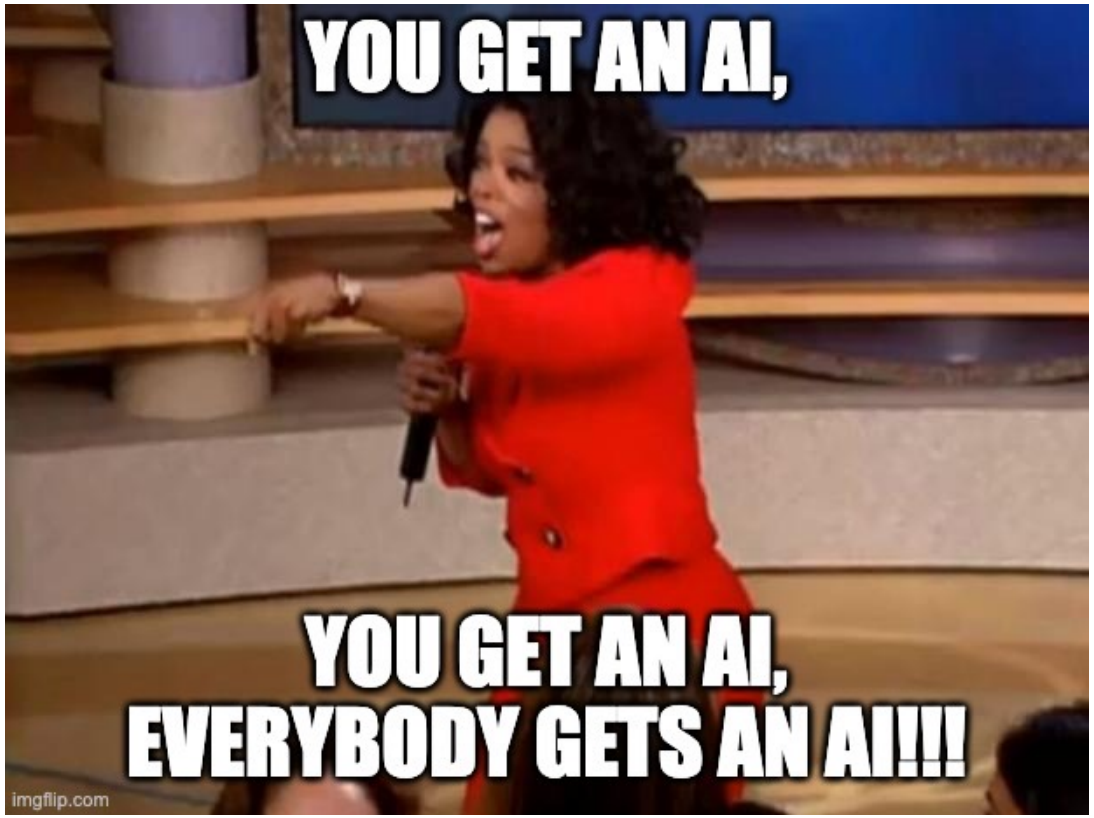
1	Introduce GAI tools early in the curriculum
2	Preserve reasoning-first GAI-assisted workflows
3	Promote graduated autonomy for GAI use in clinical care
4	Equip faculty to supervise GAI-assisted reasoning
5	Rethink clinical reasoning assessment

Source: Schaye VE, Parsons AS, Olson APJ, Abdulnour RE, Daniel M. Acad Med. 2026. doi:10.1093/acamed/wvag318

SECTION 3 · TEACHING LEARNERS ABOUT AI

Introduce Generative AI tools early in the curriculum

SCHAYE ET AL. · STRATEGY 1 OF 5



Source: Schaye VE, et al. Acad Med 2026; “Introduce GAI tools early in the curriculum” and Tables 1–2

SECTION 3 · TEACHING LEARNERS ABOUT AI

Preserve reasoning-first generative AI-assisted workflows

SCHAYE ET AL. · STRATEGY 2 OF 5

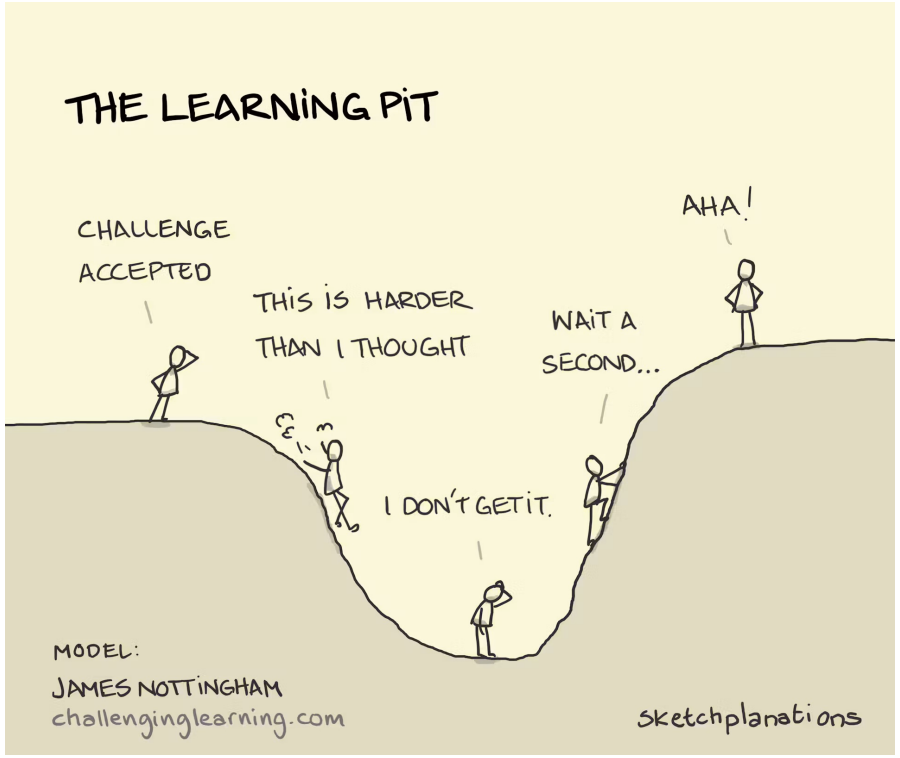
1 **Draft alone**
Without GAI, write a problem representation, differential and plan.

2 **Prompt Generative AI tool**
Ask the tool and record its version next to your own.

3 **Compare**
Set the two versions side by side.

4 **Explain**
Write a brief rationale for what you revised and what you kept.

5 **Synthesize**
Produce the final assessment and plan.



Source: Schaye VE, et al. Acad Med 2026; "Preserve reasoning-first GAI-assisted workflows" and Vignette 3

SECTION 3 · TEACHING LEARNERS ABOUT AI

Promote graduated autonomy for Generative AI use in clinical care

SCHAYE ET AL. · STRATEGY 3 OF 5

GAI use	Gate the learner passes first
On-demand consult tools	Any stage; learner shows faculty they can frame the clinical question and name the evidence needed
Ambient documentation	Residency, after demonstrated competence writing notes without assistance
GAI for evidence retrieval	Established pattern recognition; plan still contextualized and monitored unaided

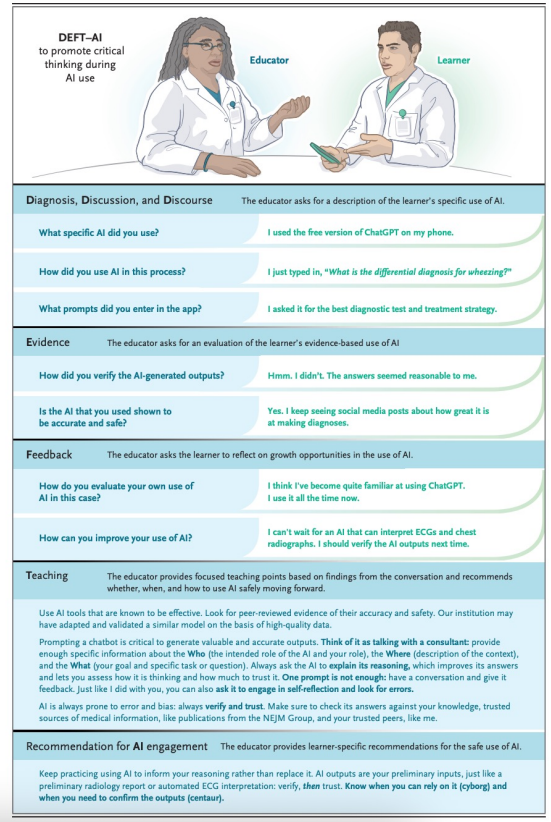
Source: Schaye VE, et al. Acad Med 2026; “Promote graduated autonomy for GAI use in clinical care”

SECTION 3 · TEACHING LEARNERS ABOUT AI

DEFT-AI: a two-minute script for the bedside

SCHAYE ET AL. · STRATEGY 4 OF 5 · EQUIP FACULTY TO SUPERVISE GAI-ASSISTED REASONING

D	**Diagnosis and discussion**	“Walk me through your thinking. Where did AI come in?”
E	**Evidence**	“What supports that answer? How did you check it?”
F	**Feedback**	“Your prompt was well framed. You took the dose without checking.”
T	**Teaching**	Deliver points on the clinical content, one on AI use.
AI	**AI engagement advice**	“Next time, build your differential first, then use AI to stress-test it.”



Framework: Abdulnour RE, et al. N Engl J Med 2025

SECTION 3 · TEACHING LEARNERS ABOUT AI

Rethink clinical reasoning assessment

SCHAYE ET AL. · STRATEGY 5 OF 5

	Without GAI	With GAI present
Purpose	Show learners can reach their own conclusions, well enough to judge GAI output	Show learners use GAI in ways that strengthen their reasoning
When and how	Checkpoints concentrated early and kept later: exams, OSCEs, virtual patients; proctored where it matters	More often as GAI enters routine care: standardized patients, chart-stimulated recall, workplace-based assessment
What is captured	Independent problem representation, differential and plan	The first draft, the prompts used, and why GAI output was accepted or rejected

Source: Schaye VE, et al. Acad Med 2026; “Rethink clinical reasoning assessment”

SECTION 3 · TEACHING LEARNERS ABOUT AI

Data-MD Curriculum

SESSION 1

AI Foundations

Pre-survey · 50 min didactic

- AI tools in clinical care today
- Key terms: ML, LLMs, generative AI
- How models are trained: fine-tuning, RAG
- Ethics: privacy, liability

FOUNDATION

SESSION 2

Applying AI to Clinical Reasoning

20 min didactic · 40 min small group

- Spot errors and hallucinations
- Name sources of bias
- Work a vignette with and without AI
- Practice prompting techniques

EVALUATE OUTPUTS

SESSION 3

Benchmarking AI for Clinical Reasoning

20 min didactic · 40 min journal club

- Human-in-the-loop vs on-the-loop
- Offloading vs automation bias
- Appraise an LLM reasoning benchmark

EVALUATE RESEARCH

SESSION 4

Critical Appraisal and the Future

35 min journal club · 15 min didactic · post-survey

- Appraise a study of LLM-written discharge summaries
- Staying current: journals, podcasts, CME
- Will AI replace doctors? Tasks first

LOOK AHEAD

DATA-MD GenAI curriculum (Data-Augmented, Technology-Assisted Medical Decision-Making) · Appraisal articles: McCoy, et al. NEJM AI 2025; Williams, et al. JAMA Intern Med 2025

SECTION 3 · TEACHING LEARNERS ABOUT AI

Data-MD Pilot Sites

OUR MULTICENTER PILOT · FIVE PARTNER INSTITUTIONS · JULY 2025–JUNE 2026

University of Michigan	University of California, Los Angeles	University of Texas at Austin	Washington University in St. Louis	Western Michigan University
Internal Medicine Residency	Internal Medicine Residency Family Medicine Residency	Pediatric Emergency Medicine Fellowship	Rheumatology Fellowship	Internal Medicine Residency

Same curriculum, adapted locally
Each site updated the curriculum through a nominal group process with its own AI experts, then assessed it with a pre/post knowledge test and a retrospective pre/post survey.

Source: DATA-MD IRB amendment, partner institutions and programs

SECTION 3 · TEACHING LEARNERS ABOUT AI

Data-MD Pilot Results

OUR MULTICENTER PILOT · PRELIMINARY RESULTS

5 health systems · 9 programs

Residents, fellows and faculty in internal, family and emergency medicine and psychiatry

KNOWLEDGE · 10 MULTIPLE-CHOICE QUESTIONS

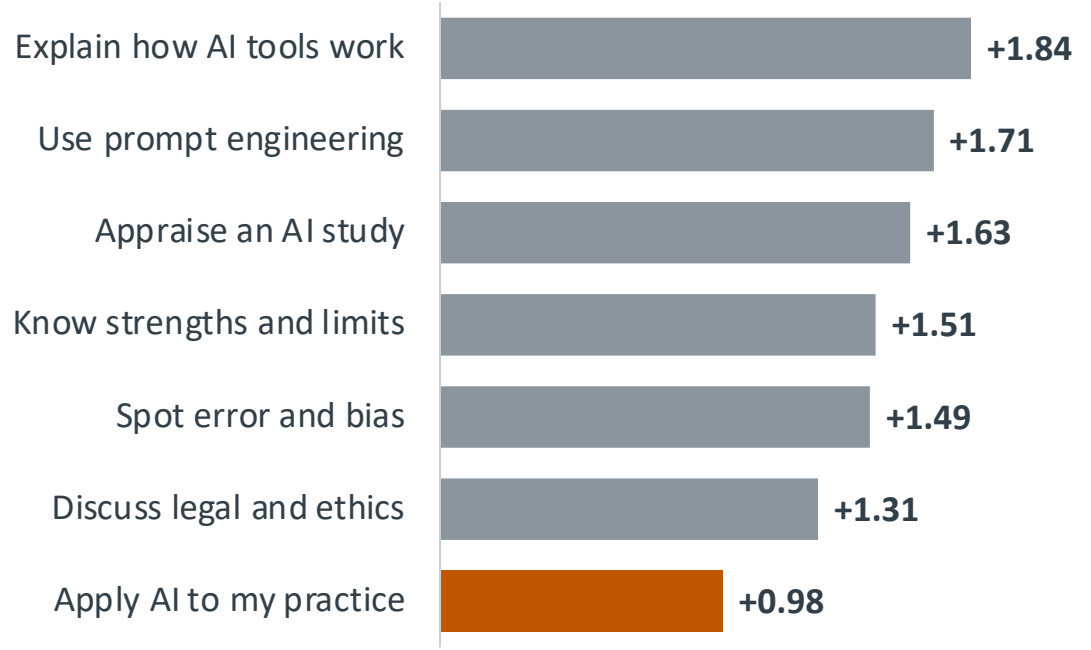
32.8% → 57.7%

pre-survey post-survey

Most valued segment: error, bias and prompt engineering (Session 2), chosen by 29%

Self-rated confidence, retrospective pre to post

Gain on a 5-point scale

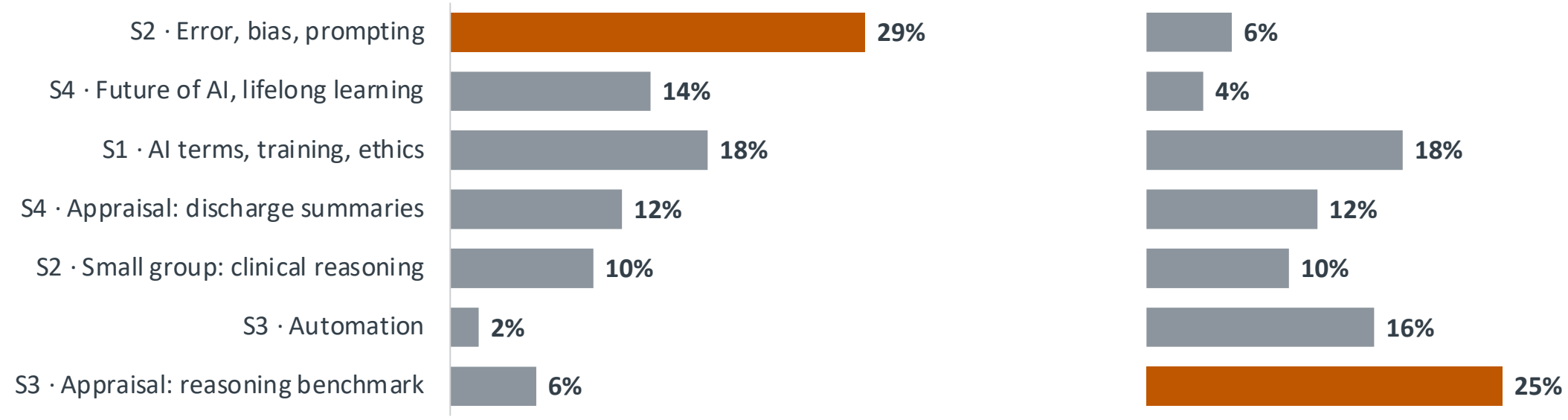


Goparaju N, et al. AI curriculum for continuing and graduate medical education: a multi-center pilot (abstract). 183 pre-survey and 51 post-survey responses; confidence self-rated on the post-survey

SECTION 3 · TEACHING LEARNERS ABOUT AI

Error and bias training was most valued; benchmarking least

PERCEIVED VALUE BY SEGMENT



Every segment was picked as most valuable by at least one learner; 10% named none.

Goparaju N, et al. AI curriculum for continuing and graduate medical education: a multi-center pilot (abstract), Table 3 · Each respondent chose one most and one least valuable segment

SECTION 4

Practical Applications and Demonstrations

SECTION 4 · PRACTICAL APPLICATIONS AND DEMONSTRATIONS

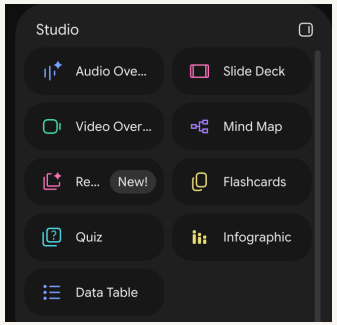
Turn a lecture into audio

Source: my 24-slide PEM lecture, *Toxicology: Bites and Stings.*

Ask for coverage, not a summary. Default audio overviews condense. Steer it to walk through every slide in order, and check it against your deck.

HOW: NOTEBOOKLM STUDIO

1. **Upload** the full lecture ppt as the only source or algorithm
2. **Customize** “Cover every slide in order. Do not condense.”
3. **Generate** an Audio Overview and download it



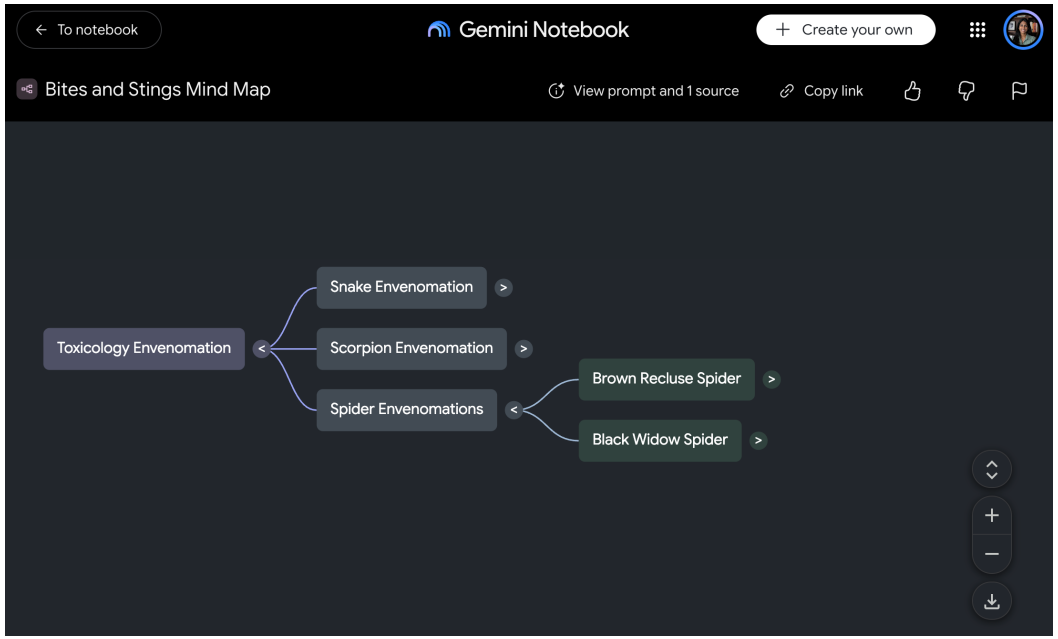
Audio Overview sits first in the Studio panel

Source: Goparaju N, Earp A. Toxicology: Envenomation (lecture) · Workflow: Konsker HB, Stanford 2026 · Wang CG, et al. Prehosp Emerg Care 2026

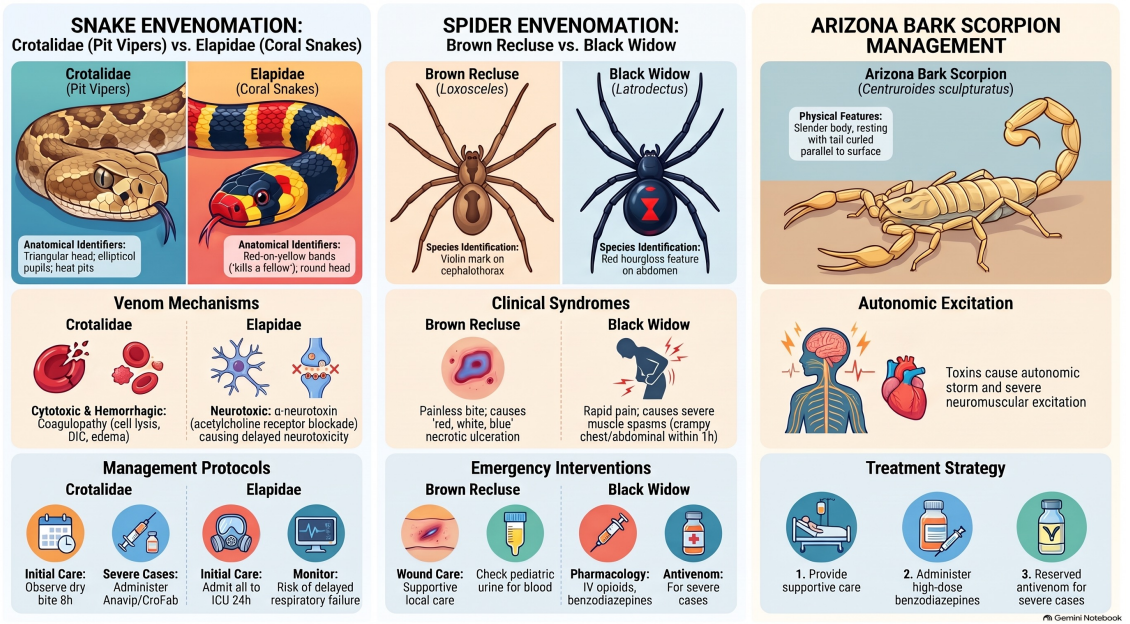
SECTION 4 · PRACTICAL APPLICATIONS AND DEMONSTRATIONS

Mind map vs. infographic (NotebookLM)

Mind map: the structure of a topic

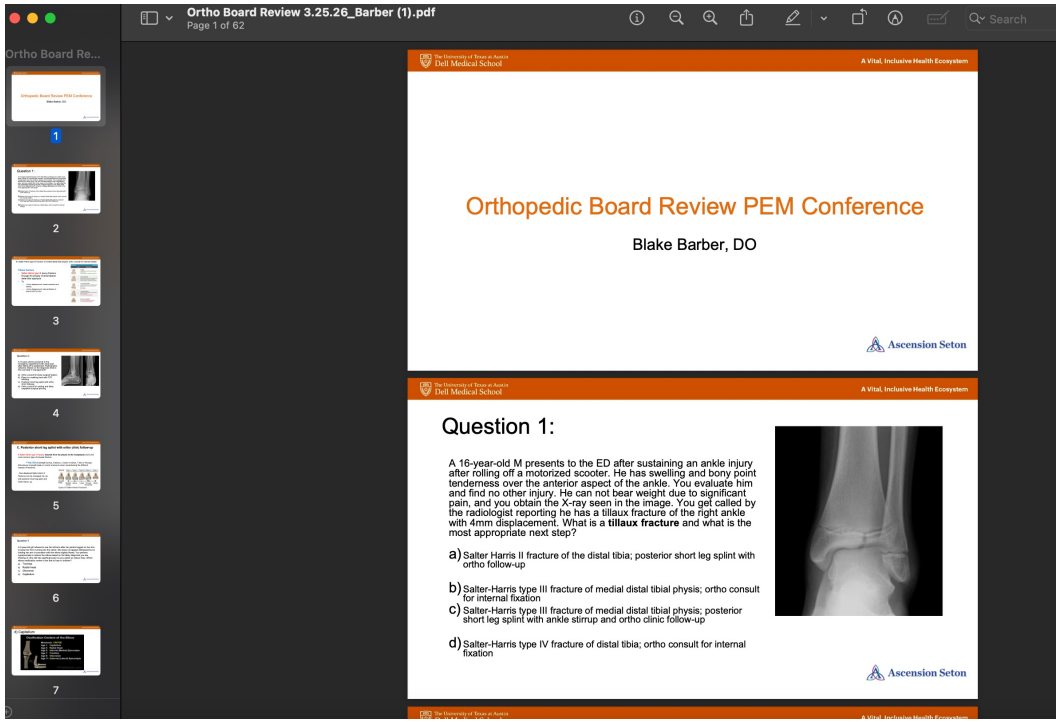


Infographic: one-page visual summary

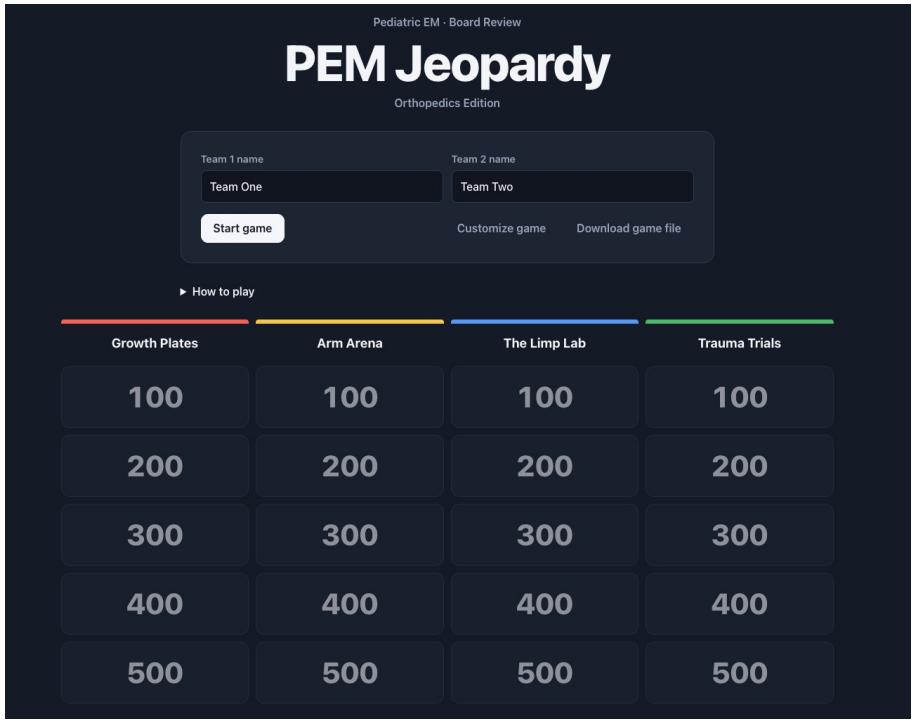


SECTION 4 · PRACTICAL APPLICATIONS AND DEMONSTRATIONS

AI-generated Board Review: Jeopardy-style



Board review PDF



AI-built Jeopardy game

SECTION 4 · PRACTICAL APPLICATIONS AND DEMONSTRATIONS

AI-generated Anki flashcards for PEM board prep

Skill will be uploaded soon

PRACTICAL APPLICATIONS · DEMONSTRATION

Simulation: Creating the sim, notes for instructor, one page summary, debrief (Claude skill)

1 · YOU TYPE

One sentence

"Build a PEM fellow sim on a 2-month-old in SVT, HR 270."

Defaults: PEM fellows, peds ED, 20 to 30 min case, 15 to 20 min debrief. Add level or emphasis if you want.

2 · THE SKILL

Same recipe every time

- Checks current guidance and doses against algorithms you place
- Follows our sim template and stage table
- Keeps patient, doses, and objectives consistent
- Flags what it can't confirm: **[VERIFY]**

3 · YOU GET

Three Word files

- **Facilitator primer:** 2-page brief so any faculty can run it
- **Sim outline:** stages, branches, scripts, labs
- **Debrief sheet:** checklist, takeaways

The point: topic to ready-to-run case in minutes, so faculty time shifts from writing to reviewing and teaching. *You stay the expert: clear every [VERIFY], adapt to local protocol, add real images.*

Demo: infant SVT package built with a custom Claude skill (pem-simulation-builder)

Example: AI-drafted SVT simulation outline

WHAT THE AI DRAFTED

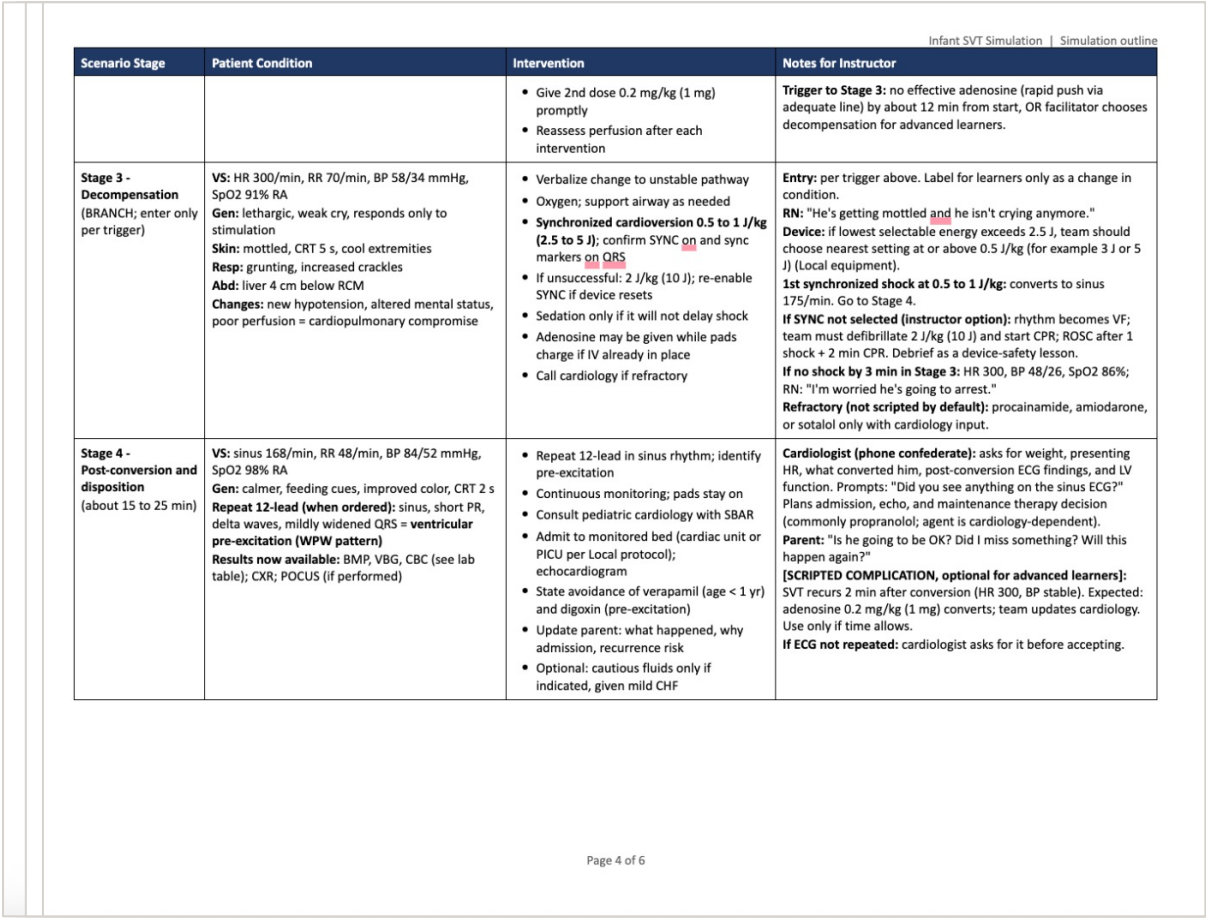
Page 1
Objectives and critical actions

Page 2
Equipment, meds, and case setup

Page 3
Stages 1–2: presentation and first doses

Page 4
Stages 3–4: decompensation and disposition

[Open the full outline (Google Doc)]

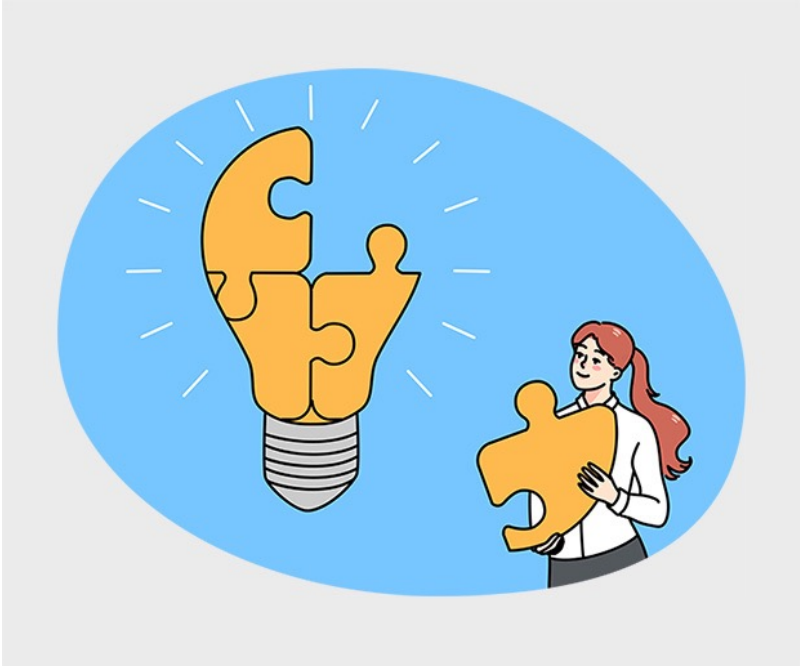


Infant SVT Simulation | Simulation outline

Scenario Stage	Patient Condition	Intervention	Notes for Instructor
		• Give 2nd dose 0.2 mg/kg (1 mg) promptly • Reassess perfusion after each intervention	**Trigger to Stage 3:** no effective adenosine (rapid push via adequate line) by about 12 min from start, OR facilitator chooses decompensation for advanced learners.
Stage 3 - Decompensation (BRANCH; enter only per trigger)	**VS:** HR 300/min, RR 70/min, BP 58/34 mmHg, SpO2 91% RA **Gen:** lethargic, weak cry, responds only to stimulation **Skin:** mottled, CRT 5 s, cool extremities **Resp:** grunting, increased crackles **Abd:** liver 4 cm below RCM **Changes:** new hypotension, altered mental status, poor perfusion = cardiopulmonary compromise	• Verbalize change to unstable pathway • Oxygen; support airway as needed • **Synchronized cardioversion 0.5 to 1 J/kg (2.5 to 5 J)**; confirm SYNC on and sync markers on QRS • If unsuccessful: 2 J/kg (10 J); re-enable SYNC if device resets • Sedation only if it will not delay shock • Adenosine may be given while pads charge if IV already in place • Call cardiology if refractory	**Entry:** per trigger above. Label for learners only as a change in condition. **RN:** "He's getting mottled and he isn't crying anymore." **Device:** if lowest selectable energy exceeds 2.5 J, team should choose nearest setting at or above 0.5 J/kg (for example 3 J or 5 J) (Local equipment). **1st synchronized shock at 0.5 to 1 J/kg:** converts to sinus 175/min. Go to Stage 4. **If SYNC not selected (instructor option):** rhythm becomes VF; team must defibrillate 2 J/kg (10 J) and start CPR; ROSC after 1 shock + 2 min CPR. Debrief as a device-safety lesson. **If no shock by 3 min in Stage 3:** HR 300, BP 48/26, SpO2 86%; RN: "I'm worried he's going to arrest." **Refractory (not scripted by default):** procainamide, amiodarone, or sotalol only with cardiology input.
Stage 4 - Post-conversion and disposition (about 15 to 25 min)	**VS:** sinus 168/min, RR 48/min, BP 84/52 mmHg, SpO2 98% RA **Gen:** calmer, feeding cues, improved color, CRT 2 s **Repeat 12-lead (when ordered):** sinus, short PR, delta waves, mildly widened QRS = **ventricular pre-excitation (WPW pattern)** **Results now available:** BMP, VBG, CBC (see lab table); CXR; POCUS (if performed)	• Repeat 12-lead in sinus rhythm; identify pre-excitation • Continuous monitoring; pads stay on • Consult pediatric cardiology with SBAR • Admit to monitored bed (cardiac unit or PICU per Local protocol); echocardiogram • State avoidance of verapamil (age < 1 yr) and digoxin (pre-excitation) • Update parent: what happened, why admission, recurrence risk • Optional: cautious fluids only if indicated, given mild CHF	**Cardiologist (phone confederate):** asks for weight, presenting HR, what converted him, post-conversion ECG findings, and LV function. Prompts: "Did you see anything on the sinus ECG?" Plans admission, echo, and maintenance therapy decision (commonly propranolol; agent is cardiology-dependent). **Parent:** "Is he going to be OK? Did I miss something? Will this happen again?" **[SCRIPTED COMPLICATION, optional for advanced learners]:** SVT recurs 2 min after conversion (HR 300, BP stable). Expected: adenosine 0.2 mg/kg (1 mg) converts; team updates cardiology. Use only if time allows. **If ECG not repeated:** cardiologist asks for it before accepting.

Page 4 of 6

Conclusion



1. **Protect human skills**
2. **Think first, then use AI**
3. **Let AI draft; you verify**

AI literacy resources
niharikagoparaju.github.io/ai-literacy

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